When Does the Chameleon Thin-Shell Approximation Fail?

Numerical Scalar-Charge Accuracy and Force-Ratio Threshold Maps

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Abstract

The long-range thin-shell law is a convenient shortcut for predicting how strongly a chameleon-screened sphere sources a fifth force, but it neither supplies an internal error estimate nor uniquely specifies how a finite-range, extended source should be represented. This study asks when that shortcut is accurate and whether its finite-range continuation can change an illustrative force-ratio threshold decision. For the inverse-power potential $V(\phi) = \Lambda^5/\phi$, I solve the full nonlinear spherical boundary-value problem at 725 logarithmically spaced points in density contrast $10^2 \le \rho_c/\rho_\infty \le 10^5$ and exterior range $10^{-3} \le m_\infty R \le 0.3$. A two-domain collocation method preserves the sharp source boundary, while two local Yukawa-tail estimators measure the numerical scalar charge. Extending the clipped long-range charge by placing it at the source surface is within 10% at 68.8% of the grid and overpredicts by as much as 62.6%; placing it at the center is within 10% at 91.3%. Thus there is no unique "canonical" finite-range continuation. An active-shell Yukawa integral constructed and benchmarked here models the radial source distribution and is within 10% at 99.7% of the same grid (0.39% median, 13.1% maximum error). The atlas uses $x_{\max} = 160$; a 14-point stratified $160 \to 240$ audit changes the charge by at most 0.056% and flips no 10%-accuracy classification. At the illustrative proxy threshold $|a_5/a_N|_\star = 10^{-2}$, the two naive continuations disagree at four grid points; neither produces a convention-independent mismatch with the numerical decision. These are dimensionless spherical-methodology results, not an experimental reach or exclusion. A conservative validity filter is also checked on 12 held-out log-midpoints.

1 Introduction

Chameleon screening was proposed to reconcile a light, matter-coupled scalar field on large scales with stringent local fifth-force tests [1, 2]. Its defining mechanical feature is environmental: the minimum and curvature of an effective potential depend on the surrounding density. A dense body can then develop a nearly constant interior field and source the scalar primarily through a thin layer near its surface. In the standard long-range asymptotic treatment this entire nonlinear boundary-value problem is replaced by one scalar-charge factor,

$$Q_{\rm LR} = \min(3\epsilon, 1), \qquad \epsilon \equiv \frac{\Delta R}{R}, \qquad m_\infty R \ll 1. \tag{1}$$

The formula is physically transparent and computationally cheap, which is why it appears throughout phenomenological estimates and experimental discussions [4]. It is not, however, an error-controlled numerical method. Its derivation assumes a small shell, an equilibrated interior, and a source much smaller than the exterior Compton wavelength. Once $m_\infty R$ is finite, the radial location of the active scalar source matters; the long-range formula alone does not prescribe it.

The validity of thin-shell and no-shell asymptotics is not a new subject. Tamaki and Tsujikawa showed explicitly that the familiar profile is recovered as $m_c R \to \infty$ and studied distinct no-shell solutions [3]. Full numerical calculations have also been performed for specific laboratory geometries. For example, Elder et al. found an approximately 20% difference between a simple analytic atom-interferometer estimate and their three-dimensional numerical calculation [7]. The narrower goal here is therefore not to claim the first test of the approximation. It is to construct an *accuracy-to-observability bridge*: a systematic error atlas across the spherical transition regime, and then propagate that uncertainty into an illustrative force-ratio decision.

The study asks three questions.

1. Where does the long-range law predict the numerical scalar charge to 10% accuracy?
2. How strongly do center-point and surface-shell finite-range continuations disagree, and can an active-shell integral model the full BVP?
3. When does that continuation ambiguity affect the answer to the proxy question "is $|a_5/a_N|$ above a chosen threshold?"

All code, data, tests, and figure-generation commands are available in the public repository Thin_Shell_Study on GitHub.

2 Physical model and complete nondimensionalization

2.1 Effective potential and density-dependent mass

For a positive scalar field with inverse-power potential and conformal matter coupling, take

$$V(\phi) = \frac{\Lambda^5}{\phi}, \qquad A(\phi) = \exp\left(\frac{\phi}{M_c}\right), \qquad M_c \equiv \frac{M_{\rm Pl}}{\beta}. \tag{2}$$

For nonrelativistic matter of density ρ, the static weak-gravity equation is

$$\nabla^2\phi = -\frac{\Lambda^5}{\phi^2} + \frac{\rho}{M_c}\exp\left(\frac{\phi}{M_c}\right). \tag{3}$$

In the laboratory regime $\phi/M_c \ll 1$, the exponential is accurately linearized. The density-dependent equilibrium and mass are then

$$\phi_{\rm min}(\rho) = \left(\frac{\Lambda^5 M_c}{\rho}\right)^{1/2}, \qquad m_{\rm eff}^2(\rho) = \frac{2\Lambda^5}{\phi_{\rm min}^3} \propto \rho^{3/2}. \tag{4}$$

Thus high density lowers the equilibrium field and shortens its Compton range $\lambda_c = m_{\rm eff}^{-1}$. This is the mechanical origin of screening.

It is important not to lose a dimensionless coupling coefficient during normalization. With

$$x = \frac{r}{R}, \quad u = \frac{\phi}{\phi_\infty}, \quad \widetilde{\rho} = \frac{\rho}{\rho_\infty}, \quad D = \frac{\phi_\infty}{M_c}, \tag{5}$$

the *unlinearized* $n = 1$ equation is

$$u'' + \frac{2}{x}u' = -\frac{A}{u^2} + B\widetilde{\rho}e^{Du}, \quad A = \frac{\Lambda^5 R^2}{\phi_\infty^3}, \quad B = \frac{\rho_\infty R^2}{M_c\phi_\infty}. \tag{6}$$

Choosing ϕ_∞ to be the exact exterior minimum gives $A = Be^D$ and $(m_\infty R)^2 = A(2+D)$. The coefficient D is therefore independent and may not simply disappear. The main scan deliberately takes the controlled limit $D \to 0$ in which the standard thin-shell formula itself is derived. Defining

$$\delta \equiv \frac{\rho_c}{\rho_\infty}, \qquad \mu \equiv m_\infty R, \tag{7}$$

reduces the spherical field equation to

$$u'' + \frac{2}{x}u' = \frac{\mu^2}{2}\left[\widetilde{\rho}(x) - u^{-2}\right], \qquad \widetilde{\rho}(x) = \begin{cases}\delta, & x < 1,\\ 1, & x > 1.\end{cases} \tag{8}$$

Regularity and environmental matching require

$$u'(0) = 0, \qquad u(\infty) = 1. \tag{9}$$

For this model,

$$u_c = \delta^{-1/2}, \qquad m_c R = \mu\delta^{3/4}. \tag{10}$$

The two inputs (δ, μ) therefore generate two diagnostic assumptions: whether the shell is thin and whether the core can settle to its minimum.

3 From field profile to scalar charge

3.1 Long-range law and finite-range continuations

Use the total body mass $M_s = 4\pi\rho_c R^3/3$ and surface potential $\Phi_N = GM_s/R$, as in the standard high-contrast convention. The shell parameter becomes

$$\epsilon = \frac{\phi_\infty - \phi_c}{6\beta M_{\rm Pl}\Phi_N} = \frac{1 - u_c}{B\delta} = \frac{2(1 - \delta^{-1/2})}{\mu^2\delta}, \qquad B = \frac{\mu^2}{2}. \tag{11}$$

The long-range prediction is equation (1). The factor 3ϵ is a first-order shell-volume result; clipping it at unity is already a heuristic transition rule. At finite μ, an additional convention is needed. In the surface-referenced tail used below, two simple continuations are

$$Q_{\rm surf} = Q_{\rm LR}, \qquad Q_{\rm ctr} = e^{-\mu}Q_{\rm LR}. \tag{12}$$

They place the effective charge at the surface and center, respectively. They coincide for $\mu \to 0$, but neither is uniquely selected by the long-range law.

A slightly more careful matching keeps the finite shell volume. If a frozen core has radius $r_s = qR$, integrating the approximately matter-dominated shell and matching to a massless exterior gives

$$\epsilon = \frac{1 - q^2}{2}, \qquad Q_{\rm MS} = 1 - q^3 = 1 - \max(1 - 2\epsilon, 0)^{3/2}. \tag{13}$$

This agrees with 3ϵ at first order but reaches the unscreened limit smoothly at $\epsilon = 1/2$.

The standard Yukawa integral of a uniform extended source gives, in this surface-referenced convention,

$$Q_{\rm lin}(\delta, \mu) = \left(1 - \delta^{-1}\right)\frac{3e^{-\mu}}{\mu^3}\left(\mu\cosh\mu - \sinh\mu\right), \tag{14}$$

where $1 - \delta^{-1}$ subtracts the uniform ambient source. Relative to a center-referenced point source, the corresponding form factor is $3g(\mu)/\mu^3 > 1$; "finite-range suppression" is therefore not an invariant description. Applying the standard radial Yukawa weight to a shell with the massless-matching radius motivates the project-specific ansatz

$$g(z) = z\cosh z - \sinh z, \tag{15}$$

$$Q_{\rm AS} = \frac{3e^{-\mu}}{\mu^3}\left[g(\mu) - g(\mu q)\right], \qquad q = \sqrt{\max(1 - 2\epsilon, 0)}. \tag{16}$$

It reduces to equation (13) as $\mu \to 0$ and to 3ϵ when both ϵ and μ are small. In the fully active limit it has the standard uniform-sphere radial factor but omits the small ambient subtraction. Equation (16) is constructed and benchmarked here; it is neither a literature-priority claim nor an assumed property of the nonlinear solution. It does not self-consistently rematch the core radius at finite μ.

3.2 Numerical charge and force ratio

In the exterior linear regime the numerical field is written as

$$u(x) = 1 - sQ_{\rm num}\frac{e^{-\mu(x-1)}}{x}, \qquad s \equiv \frac{B\delta}{3}. \tag{17}$$

This defines the dimensionless scalar charge relative to the total body mass; the ambient contribution cancels dynamically in the BVP. Two local estimators follow:

$$Q_A(x) = \frac{x[1-u(x)]e^{\mu(x-1)}}{s}, \tag{18}$$

$$Q_\nabla(x) = \frac{x^2u'(x)e^{\mu(x-1)}}{s(1+\mu x)}. \tag{19}$$

The reported $Q_{\rm num}$ is the robust median of Q_A over an exterior window where the exact source term differs from its linearization by less than 2%; the gradient estimator is an independent diagnostic.

For a test body of scalar charge Q_t at $x_p = r_p/R$, the idealized force ratio is

$$\alpha_5 \equiv \left|\frac{a_5}{a_N}\right| = 2\beta^2 Q_s Q_t (1+\mu x_p)e^{-\mu(x_p-1)}. \tag{20}$$

An unscreened atom has $Q_t \simeq 1$, whereas a macroscopic test body can have $Q_t < 1$ [5]. This distinction concerns physical response only: atom interferometers and bulk force sensors have different noise, systematics, and geometries, so Q_t alone is not an instrument ranking.

4 Numerical method and verification

4.1 Two-domain boundary-value solver

The density is discontinuous at $x = 1$. Smoothing that step would introduce an artificial interface width that can be larger than a genuine thin shell. Instead, the interior $0 \le x \le 1$ and exterior $1 \le x \le x_{\rm max}$ are mapped to two copies of a common collocation coordinate. The field and its radial derivative are matched exactly at the surface. At the finite outer boundary, the Yukawa Robin condition

$$u'(x_{\rm max}) = \left(\mu + \frac{1}{x_{\rm max}}\right)[1-u(x_{\rm max})] \tag{21}$$

replaces a less accurate finite-radius Dirichlet condition.

To condition the scalar charge directly, define

$$x(u-1) = sv. \tag{22}$$

Then an exterior solution has $v = -Q_{\rm num}e^{-\mu(x-1)}$, keeping the fitted unknown of order unity even when $u-1$ is tiny. Strongly screened points use this linear field representation. Transition and unscreened points use the positive representation $u = \exp(sv/x)$, which avoids nonphysical negative Newton iterates. Both representations solve the same equation (8); at a cross-check point near the switching region their charges differ by only 2.3×10^{-9} fractionally.

SciPy's fourth-order collocation BVP algorithm with adaptive mesh refinement is used at a nominal tolerance 2×10^{-4} for the atlas and as tight as 2×10^{-7} for verification. The scan covers

$$10^2 \le \delta \le 10^5 \quad (25 \text{ points}), \qquad 10^{-3} \le \mu \le 0.3 \quad (29 \text{ points}), \tag{23}$$

for 725 logarithmically spaced parameter combinations. All points converged. The atlas uses $x_{\rm max} = 160$; $x_{\rm max} = 240$ is reserved for the domain check. The grid is a numerical design, not a probability distribution; percentages below always mean fractions of these sampled points.

4.2 Verification results

Figure 1 summarizes three independent checks. First, at a weak density contrast $\delta = 1.01$, where the nonlinear problem must approach the uniform linear Yukawa sphere, the maximum error over $\mu = (0.01, 0.03, 0.1, 0.3)$ is 2.8×10^{-5}. Second, at the transition benchmark $(\delta, \mu) = (10^5, 0.01)$, increasing $x_{\max}$ from 160 to 240 changes Q by 0.0528%, while tightening the solver tolerance by two decades changes it by 5.6×10^{-7}. Third, the amplitude and gradient estimators agree to 0.0339% in the asymptotic window. Across the full atlas their largest disagreement is 0.0481%. Numerical extraction uncertainty is therefore much smaller than the 10% physical-approximation target. A separate 14-point stratified audit includes all four grid corners, the two displayed benchmarks, and points nearest the surface and active-shell 10% error boundaries. Its largest $160 \to 240$ charge change is 0.0560%, and none of the surface, center, or active-shell 10% classifications changes. This directly checks the decision boundary, although it is not a point-by-point convergence proof for the entire atlas.

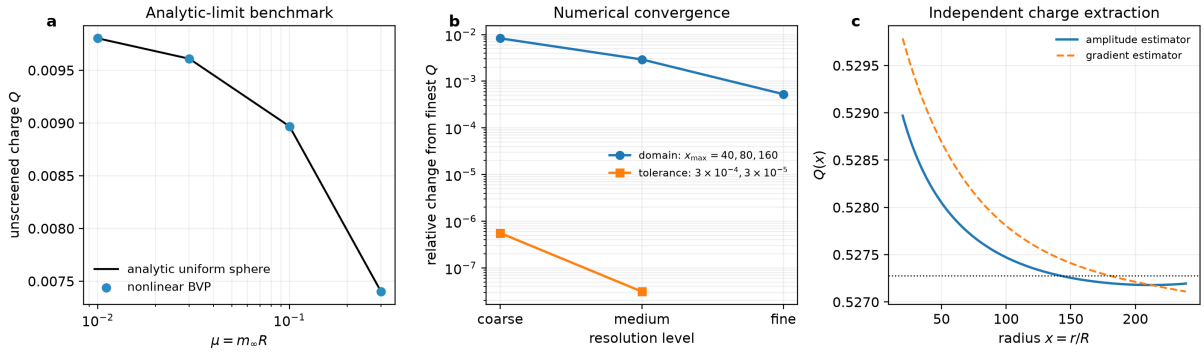


Figure 1: Numerical verification. (a) The nonlinear BVP reproduces the known uniform-sphere Yukawa amplitude in the weak-source limit. (b) Outer-domain and collocation-tolerance convergence. (c) Independent exterior amplitude and gradient charge estimators approach the same plateau.

5 Results

5.1 Three screening regimes

Figure 2 shows representative unscreened, transition, and screened solutions. The unscreened field varies throughout the source and has $Q_{\text{num}} = 0.885$. The transition case has an interior close to its density minimum but a visibly extended surface transition, with $Q_{\text{num}} = 0.499$. The strongly screened source is almost frozen inside and has $Q_{\text{num}} = 5.47 \times 10^{-3}$. In every case the two exterior charge estimators form the same plateau, demonstrating that the plotted charge is measured from the full solution rather than inserted through a thin-shell assumption.

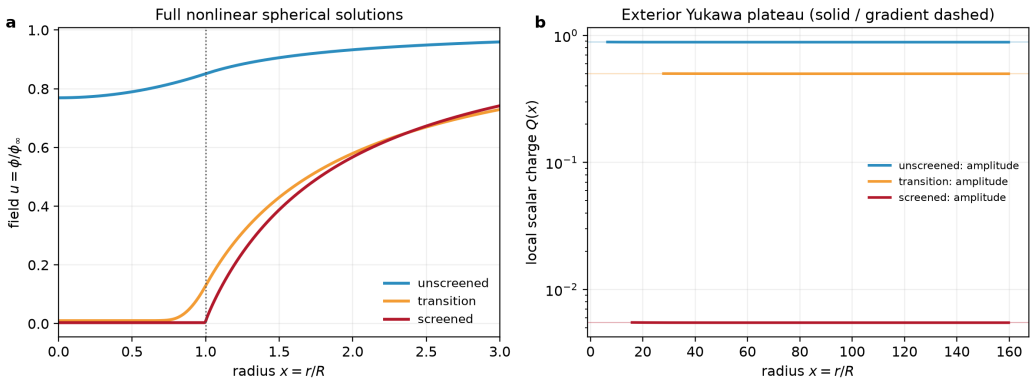


Figure 2: Full nonlinear field profiles and charge extraction. The vertical dotted line marks the source surface. Solid and dashed curves in the right panel are the amplitude and gradient estimators of equation (19).

5.2 Accuracy atlas and failure conditions

Figure 3 is the main approximation-validity result. Color represents the signed logarithmic error: red means the approximation overpredicts the numerical charge. The black curve is the 10% error boundary. The canonical long-range charge, when continued from the surface, is accurate in the small-μ portion but fails in a broad transition band. The center-point continuation is often better but develops both positive and negative bias. The active-shell integral models the radial source distribution and removes most of the trend.

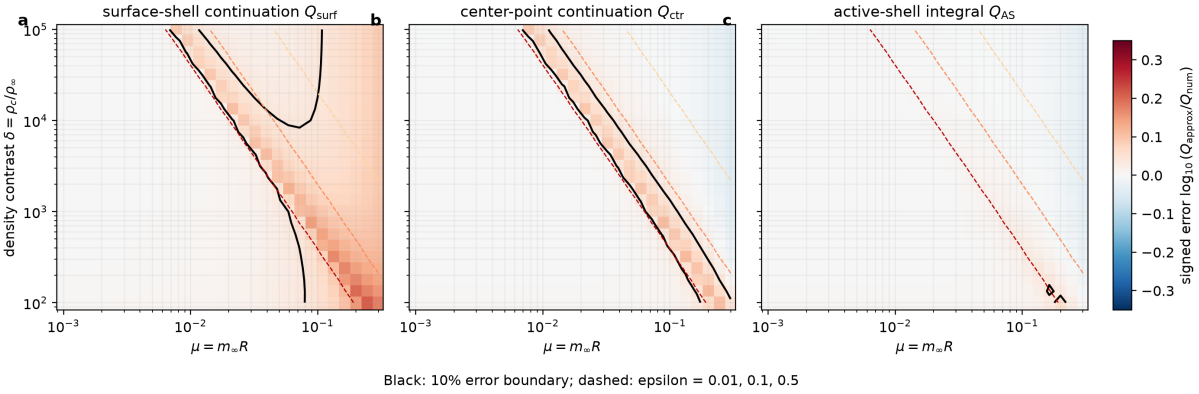


Figure 3: Signed scalar-charge error for surface-shell and center-point continuations of the long-range law, and for the active-shell integral. Black curves mark 10% relative error; dashed curves show $\epsilon = 0.01, 0.1, 0.5$. Finite-$\mu$ continuation is not fixed by ϵ alone.

Table 1: Error statistics over all 725 logarithmic grid points.

Approximation	Median error	90th percentile	Maximum	Within 10%
Surface shell $Q_{\rm surf}$	2.78%	23.0%	62.6%	68.8%
Center point $Q_{\rm ctr}$	0.566%	8.89%	30.1%	91.3%
Finite shell only $Q_{\rm MS}$	2.18%	18.0%	37.6%	77.5%
Active-shell integral $Q_{\rm AS}$	0.391%	3.66%	13.1%	99.7%

The surface-shell continuation is larger than $Q_{\rm num}$ at every sampled point. Its largest error, 62.6%, occurs at $(\delta, \mu) = (100, 0.245)$ with $\epsilon = 0.301$ and $m_c R = 7.74$: neither the small-

shell nor heavy-core assumption is parametrically secure, and finite range is appreciable. More surprisingly, imposing only the familiar conditions $\epsilon < 0.1$ and $m_c R > 10$ is insufficient: only 32.8% of the 180 sampled points satisfying those two conditions meet the 10% target because μ can still be too large. A conservative filter selected from this grid is

$$\boxed{\epsilon < 0.1, \qquad m_c R > 10, \qquad m_\infty R < 0.035.} \tag{24}$$

All 17 atlas points behind this filter are within 10%, with an 8.09% maximum error. I then test 12 logarithmic midpoint combinations excluded from the atlas; all remain within 10%, with an 8.15% maximum. This is an empirically validated filter for the stated domain, not a universal theorem. Its significance is conceptual: a small shell and a heavy core do not by themselves justify a finite-range continuation.

Figure 4 makes the separation quantitative. The two naive continuations bracket different parts of the grid. The active-shell integral follows the numerical diagonal closely; its two points outside 10% lie in the low-contrast, intermediate-screening corner, where a frozen-core shell ansatz is least reliable.

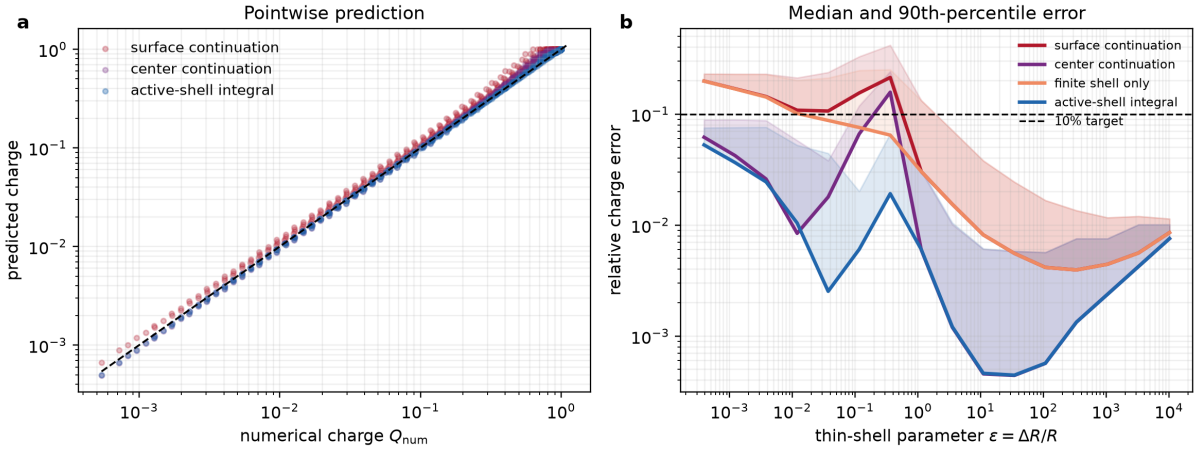


Figure 4: Pointwise predictions and binned errors. The filled bands in the right panel span the median to the 90th percentile within bins of ϵ. The non-monotonic continuation error confirms that ϵ alone is not a complete validity variable.

5.3 Illustrative force-ratio threshold map

To propagate charge uncertainty into a dimensionless proxy decision, set $\beta = 1$, $Q_t = 1$, and $x_p = 1.5$ in equation (20). The left panel of figure 5 shows $|a_5/a_N|$ for a hypothetical unscreened probe. It is not an apparatus reach or discovery forecast: (δ, μ) does not fix the absolute Newtonian acceleration, and no detector noise, source scale, or systematic uncertainty is included.

For a transparent stress test, choose $\alpha_\star = |a_5/a_N|_\star = 10^{-2}$. A point is labeled "above" when its predicted ratio exceeds this illustrative threshold. The surface and center continuations disagree at four grid points. At every one of them the numerical classification agrees with one continuation, so there is no continuation-independent mismatch. Among seven density rows with an in-grid crossing, the median numerical-contour displacement is 0.032 dex for the surface continuation and 0.0066 dex for the center continuation. The third panel shows the larger of the two naive-continuation errors in units of $\alpha_\star$.

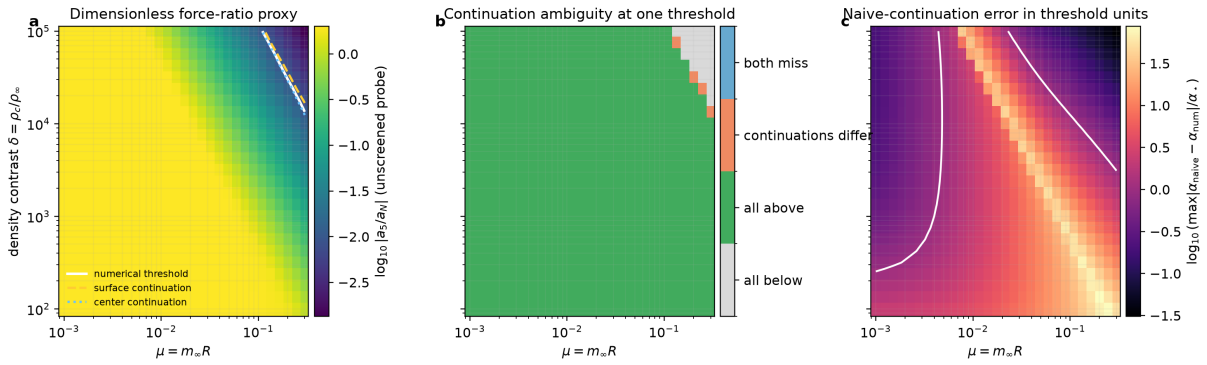


Figure 5: Accuracy-to-observability proxy. (a) Numerical fifth-force ratio for an unscreened probe, with numerical, surface-shell, and center-point threshold contours. (b) Points where the two finite-range continuations agree or differ. (c) The larger naive-continuation error in units of the illustrative threshold $\alpha_\star = 10^{-2}$; the white contour marks unit error.

For a real past null experiment, the appropriate quantity would instead be a reach ratio $\mathcal{R} = |a_5|/a_{\rm lim}$: $\mathcal{R} > 1$ would mean that the predicted signal exceeds the published limit only if geometry, signal template, nuisance treatment, and confidence construction are matched; it does not mean that a force was discovered. For a future design one would use $\mathcal{D} = |\Delta a_5|/\sigma_{\Delta a}$ and state the statistical and systematic assumptions. Atom-interferometer experiments [6, 9] and levitated bulk sensors [10] demonstrate that both probe types can be powerful despite very different screening and noise mechanisms.

6 Discussion

6.1 What the result adds

The project now follows a conventional research logic.

Problem.
The long-range charge shortcut has no error bar through the screening transition and no unique finite-range continuation.

Proposed method.
Solve the nonlinear spherical BVP, measure an exterior charge without assuming a shell, compare two explicit continuations, and propagate their ambiguity into a threshold proxy.

Result.
Surface and center continuations differ substantially at finite range; an active-shell radial integral models most of the numerical trend.

Limited contribution.
The atlas supplies a held-out-checked validity filter, and equation (16) works at the 10% level for 99.7% of this grid.

The novelty is therefore not "discovering screening" and not claiming the first thin-shell test. It is the systematic connection between approximation error and an observability proxy, together with a reproducible benchmark of an active-shell ansatz built from the standard Yukawa radial integral.

6.2 Relation to laboratory calculations

Laboratory predictions contain additional layers absent here. Chamber walls can set the environmental field and change the force substantially; numerical four-region studies have shown that geometry and walls matter [8]. Elder et al.'s full three-dimensional calculation found that the simple analytic prediction for a specific atom-interferometer geometry was roughly 20% high

[7]. This establishes the direction of the shift for that configuration only; walls and geometry can alter predictions in either direction. Those percentages are not directly comparable to table 1, because the present study isolates a single uniform sphere and deliberately excludes chamber geometry.

The contemporary status of the underlying dark-energy model must also be stated honestly. Yin et al. report that their levitated force-sensor experiment rules out the basic cosmologically viable chameleon model in the parameter space they study [10]. The present calculation does not reopen that claim. It treats the $n = 1$ chameleon as a clean nonlinear mechanics laboratory for studying screening approximations and experimental-design bias.

6.3 Limitations and next tests

The conclusions are conditional on a static, isolated, uniform spherical source; $n = 1$; weak conformal coupling $D \to 0$; and the sampled (δ, μ) range. They do not include:

- chamber walls, finite vacuum regions, source modulation, or nonspherical geometries;
- a numerical scalar charge for the test body or device-specific noise;
- other inverse-power indices, relativistic compactness, or time-dependent fields;
- a physical (Λ, M_c) exclusion map.

A direct next project would solve a separate empty-chamber BVP to determine the local environmental field, add source and wall regions, and compare atom and bulk probes using their own Q_t and σ_a. Keeping the isolated-sphere atlas separate is essential: otherwise intrinsic thin-shell error cannot be distinguished from geometry error.

7 Conclusion

The long-range thin-shell approximation is useful, but "ϵ is small" does not define a unique finite-range prediction. Across 725 nonlinear spherical solutions, the surface-shell continuation is within 10% at 68.8% of the grid and can overpredict by 62.6%; the center-point continuation reaches 91.3%. The conservative filter $\epsilon < 0.1$, $m_c R > 10$, and $m_\infty R < 0.035$ succeeds on 17 atlas points and 12 held-out midpoints. An active-shell Yukawa integral reaches 10% accuracy at 99.7% of the grid, with a 0.39% median error. In a 14-point stratified $160 \to 240$ audit, the maximum charge change is 0.056% and no 10%-accuracy classification flips. At one illustrative threshold the two naive continuations disagree at four points, while neither disagrees with the numerical result independently of convention. The scoped research result is therefore sharper than a claimed experimental forecast: the active radial distribution, not ϵ alone, controls how a screened source should be continued to finite range.

A Reproducibility

From the repository root, the complete scan, all figures, and metrics are regenerated with

```
python -m Thin_Shell_Study.run_study --workers 4 --force
python -m unittest discover -v
```

The principal files are:

- `Thin_Shell_Study/thin_shell_model.py`: BVP and charge model;
- `Thin_Shell_Study/run_study.py`: scan, verification, and plots;
- `Thin_Shell_Study/results/thin_shell_scan.csv`: all 725 points;
- `Thin_Shell_Study/results/gate_validation_midpoints.csv`: held-out filter checks;
- `Thin_Shell_Study/results/stratified_domain_audit.csv`: longer-domain spot checks;
- `Thin_Shell_Study/results/study_metrics.json`: reported numbers;
- `tests/test_thin_shell_study.py`: analytic and numerical tests.

The software environment used here is Python 3 with NumPy, SciPy, and Matplotlib as listed in the repository's `requirements.txt`.

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